

Jerry's Engineering Projects

Personal Website: iamjerryhu.org

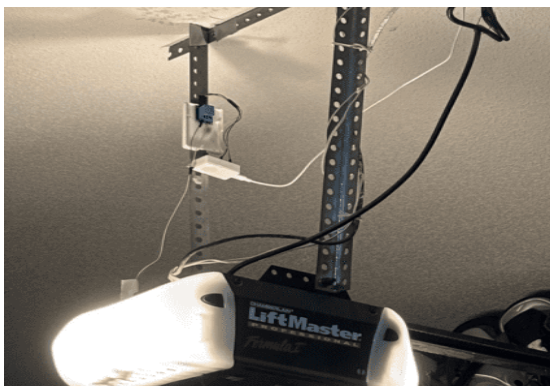
GitHub Profile: github.com/AccessRetrieved

Constellation Networking: constellationhub.org

Over the past few years, I've pursued my passion for engineering nonstop, driven by the urge to solve problems. Below is a selection of projects I've built and deployed to address problems around me.

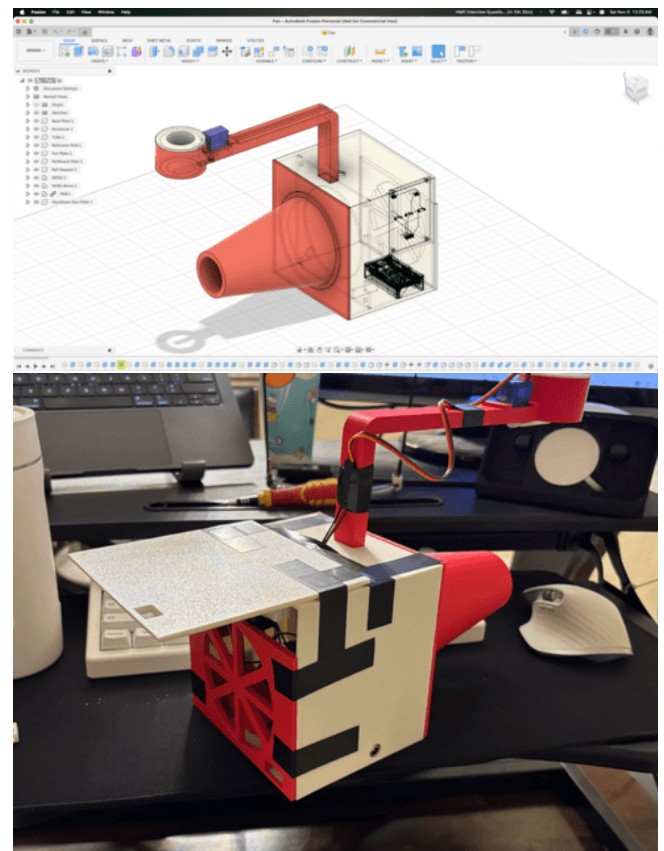
Garage Door Controller (June 2025)

Because our RF garage door remote often doesn't work, my dad purchased a MyQ device from the popular brand Chamberlain. However, we live in an area with poor cell coverage, which makes the MyQ app unreliable—and they charge \$10/month just to integrate it with our car. So, to bypass MyQ's preposterous subscription fee and the unreliable mobile connectivity, I built my own garage automation system using an ESP32 microcontroller. I designed a PCB board that allowed me to connect the ESP32 to the garage motor's motherboard via a relay and programmed it to detect my phone acting as a Bluetooth Low Energy (BLE) beacon. When I drive near the driveway, the ESP32 measures the signal strength (RSSI) to estimate the distance. Once in range, it fetches my phone's connected Bluetooth devices. If the car's Bluetooth ID appears on the list—confirming that I'm both moving towards the house and inside the car—the ESP32 activates the relay to open the garage.



Custom Fan for my IB Physics IA (Sept. 2025)

Due to the experimental design of my IB Physics IA, I needed a fan with at least 10 discrete levels of airflow—something virtually impossible to find on the market. So, I engineered my own. I used an ESP32 microcontroller to control a brushless DC fan via a Pulse Width Modulation (PWM) signal. Using an ESP32 and Raspberry Pi Zero running MicroPython, I wrote a script to programmatically control the fan speed. I later integrated a servo motor to automate a ball-drop mechanism to collect data autonomously during trials. I designed the model in *AutoDesk Fusion 360*, the PCB board in *Kicad*, and 3D-printed the housings and assembled the fan myself.



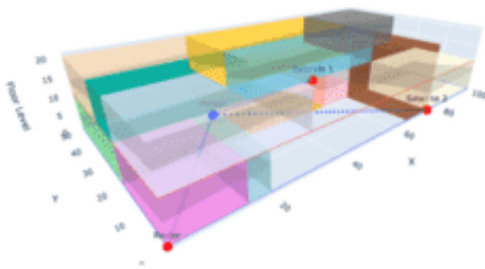
3D Model Demonstration:

<https://tinyurl.com/jerryhmcprojects>

WiFi Triangulation Algorithm (Oct. 2024)

After watching a movie in which police tracked criminals using cell tower triangulation, I decided to recreate a version of it at home. My home is outfitted with a mesh WiFi system (a main router with three satellite nodes), which became the basis of my tracking system. I wrote a Python library that allowed my code to log into the router's admin dashboard and pull the connected device data in JSON format, including each device's RSSI (Received Signal Strength Indicator) values from each of the three nodes. Given that the main router and two of the nodes are installed on the first floor and one node is installed on the second floor, I wrote a vector triangulation algorithm that sampled the RSSI values of known devices in specific rooms and created a labeled dataset, which was used to train a machine learning model to predict room-level location based on signal patterns.

Then, I created an interface that allowed me to visualize the vector algorithm in a 3D diagram of my house:



Custom Eyeglasses (November 2025)

My dad gave me a pair of red-tinted glasses to improve my sleep quality by filtering out blue light, but I couldn't use them, because I remove my contacts at night and wear prescription glasses. So, I disassembled the red lenses and designed a 3D-printed clip that attaches to my regular glasses via screws. This simple yet practical adapter lets me preserve my prescription vision while filtering blue light.

Motion Sensor (April 2025)

To make my home "smart," I needed motion sensors for presence detection—sensors that would allow my code to automatically control lights, speakers, and TVs based on room occupancy. However, commercial motion sensors are overpriced at around \$15 each, making purchasing sensors for every room impractical. So, I engineered my own solution. I used an ESP32 microcontroller paired with a PIR (passive infrared) sensor sourced from AliExpress. I designed custom PCB boards and hand-soldered each unit, then designed and 3D printed enclosures to house the electronics. Each sensor connects wirelessly to my home automation system via a Mosquito MQTT broker, continuously reporting occupancy status to my smart home server. This system of DIY sensors now automatically controls devices in my house, turning them off when unused for a set period and turning lights on when someone walks into the room.

